

Memory consolidation:

The anatomical locus and psychological character of memory both change over time.

Our understanding of this has been heavily influenced by human neurology patients with cognitive impairments from brain damage

Watch video excerpt from patient E.P. (Youtube). E.P. has bilateral hippocampal damage, cannot form new explicit memories, and has amnesia for recent memories. Very old memories appear to be intact.

Standard theory of memory consolidation:

- Memories are initially stored in hippocampus.
- Over time, memories consolidate to distributed neocortical networks for long-term storage.

Hippocampal replay:

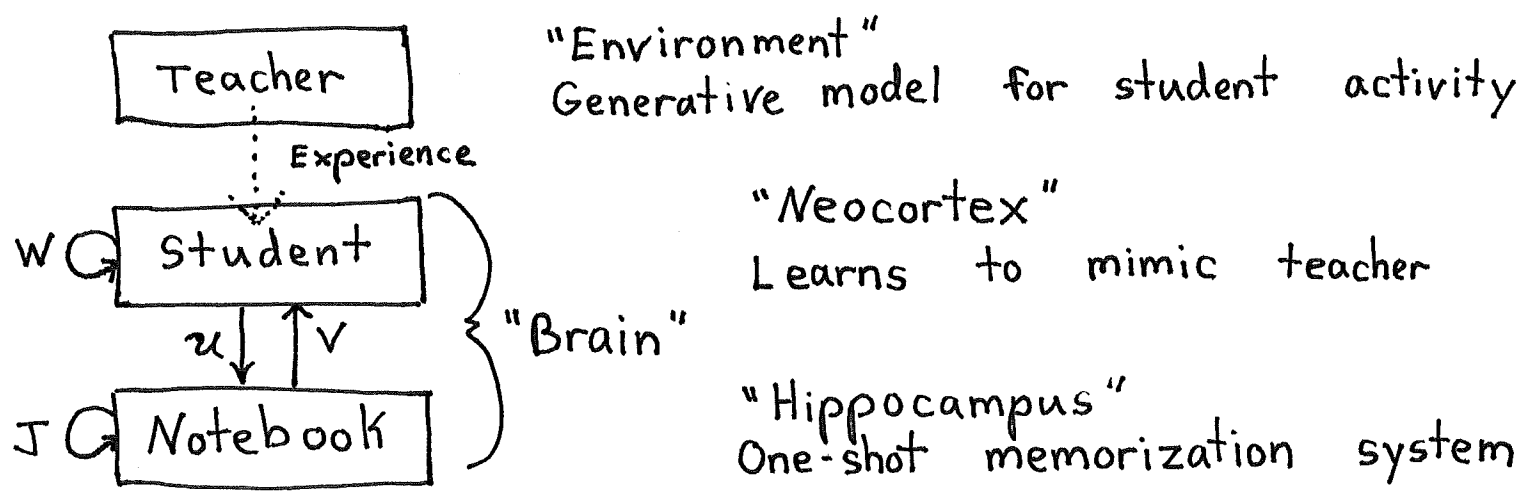
- "Place cells" report where animal is in environment.
- During rest, "place cells" replay trajectories taken through environment, compressed in time.
- This reactivates neocortical activity and could drive plasticity underlying systems consolidation.

Memory quality:

- We remember recent memories more vividly.
- Older memories are more gist like.
- Hippocampus: detailed episodic memories
- Neocortex: generalized semantic memories

Teacher - Student - Notebook Framework:

My collaborators and I recently introduced a neural network framework for modeling systems consolidation.

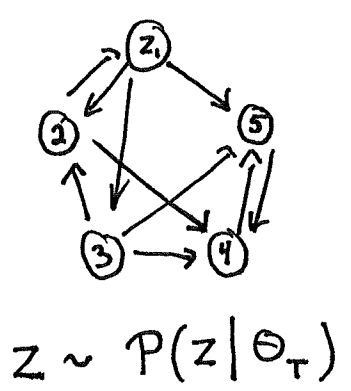


Modeling experiences:

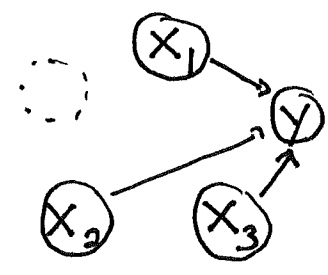
$$z \sim \mathcal{P}(z | \theta_T)$$

ND student activity Generative model
 θ_T : Teacher parameters

Goal of system: Build predictive model of student activity through learning



single Relationship



$$\hat{y} = \mathcal{F}(x | \theta_s)$$

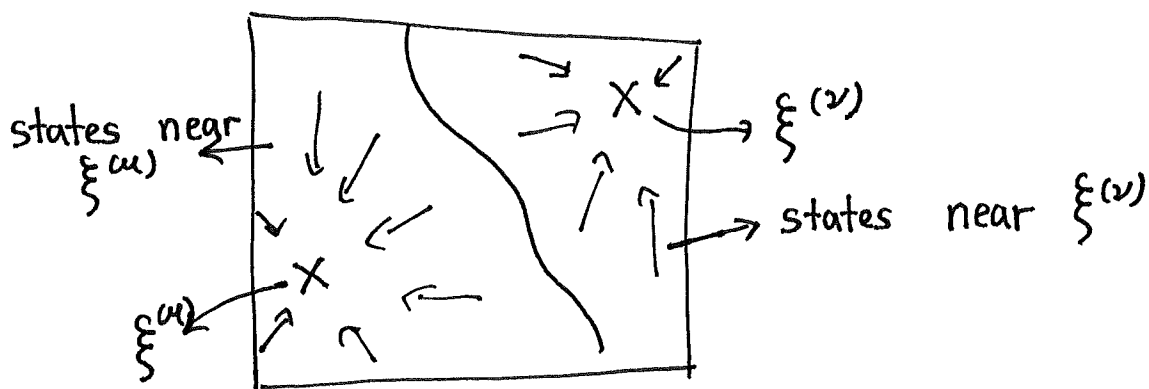
Predicted activity Student parameters

The notebook aids student learning by providing it with memory.

Want to store: $\underbrace{\{Z^{(\mu)}\}}_{\text{student activity during encoding}}, \underbrace{\{\xi^{(\mu)}\}}_{\text{notebook activity during encoding}}, \mu=1, \dots, P$

Achieved by modeling notebook as an auto-associative Hopfield model (last lecture). [Buhmann et al variety]

- Pattern separation: $\xi^{(\mu)}$ is random and independent of $Z^{(\mu)}$. This reduces interference between memories and facilitates Hebbian learning of J .
- Pattern completion: Attractor dynamics drive notebook into a memorized pattern, $\xi^{(\mu)}$, through J .



- Hebbian learning between student & notebook permits student reactivation by notebook

$$\tilde{Z}_{i\mu} = \sum_{j=1}^M V_{ij} \xi_{j\mu}$$

$$V_{ij} = \frac{1}{Ma(1-a)} \sum_{\mu=1}^P Z_{i\mu} (\xi_{j\mu} - a)$$

$$\Delta V_{ij}^{(\mu)} = \frac{1}{Ma(1-a)} Z_{i\mu} (\xi_{j\mu} - a)$$

and notebook seeding by student.

Linear model: I'll present results from an analytically tractable linear model of this framework.

$$\text{Teacher: } y = \sum_{i=1}^N \underbrace{\bar{W}_i}_{\text{Teacher weights}} x_i + \underbrace{\epsilon}_{\text{Noise}}$$

$$\text{Student: } \hat{y} = \sum_{i=1}^N \underbrace{W_i}_{\text{Student weights}} x_i$$

Memory assessed with P stored patterns.

$$e_m^2 = \frac{1}{P} \sum_{\mu=1}^P (y_{\mu} - \hat{y}_{\mu})^2 = \frac{1}{P} \sum_{\mu=1}^P (y_{\mu} - \sum_{i=1}^N W_i \underbrace{\tilde{x}_{i\mu}}_{\substack{\text{Recalled } x_{i\mu} \\ \approx x_{i\mu}}})^2$$

Generalization assessed over all possibilities

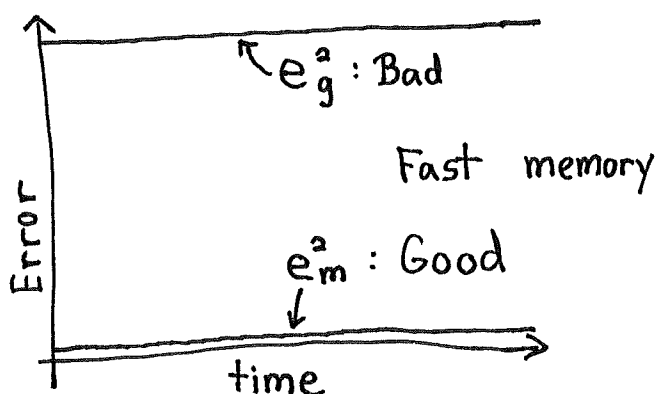
$$e_g^2 = \underbrace{\langle (y - \hat{y})^2 \rangle}_{\text{Average over } P(z|\theta_T)} = \langle (y - \sum_{i=1}^N W_i x_i)^2 \rangle$$

Standard model of systems consolidation. Adjust W to minimize e_m^2 and transfer memories.

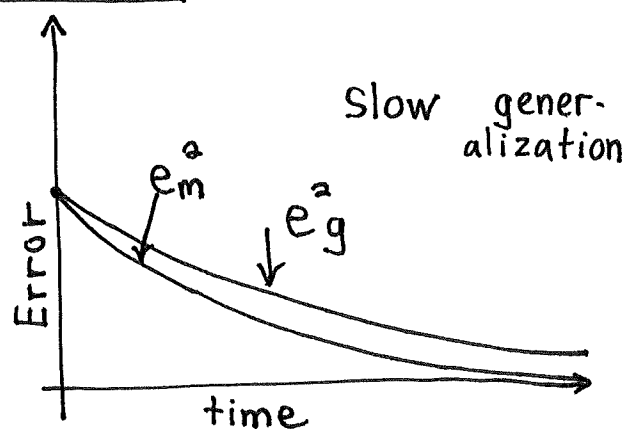
$$\tau \frac{dW_i}{dt} = - \frac{\partial}{\partial W_i} e_m^2 = \frac{2}{P} \sum_{\mu=1}^P e_{\mu} x_{i\mu}$$

First assume $\epsilon = 0$. Reveals complementary learning systems (idea from McClelland, McNaughton, O'Reilly)

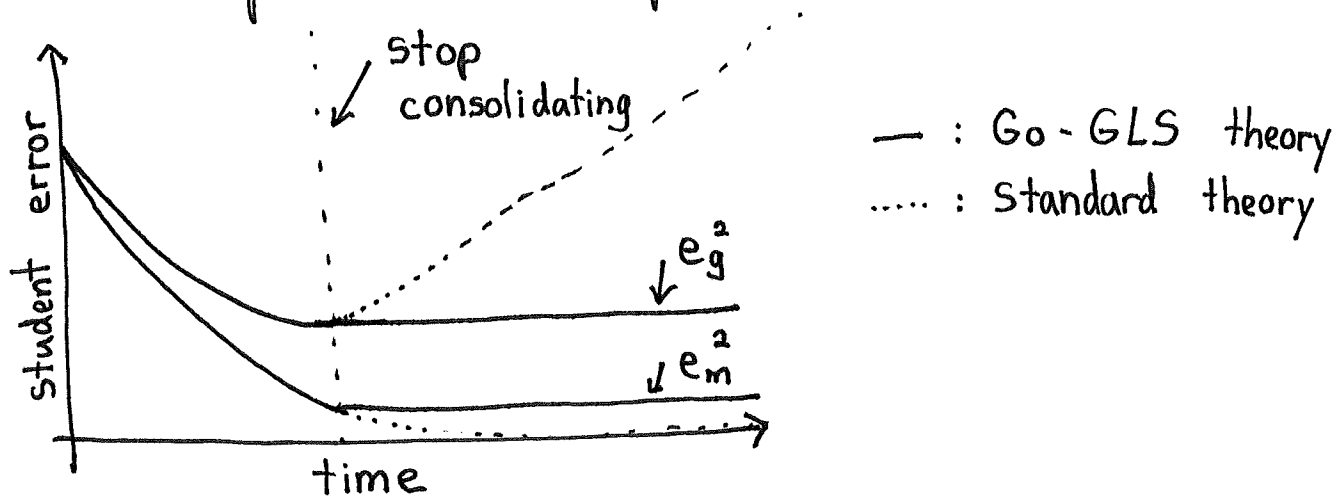
Notebook



Student



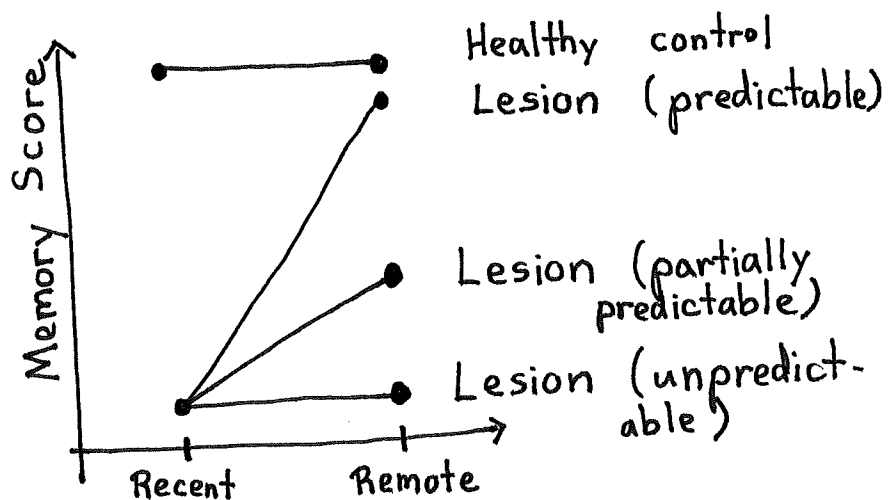
If $\epsilon \neq 0$, maintaining this generalization benefit requires student learning to stop before it overfits unpredictable components of experience.



Generalization-optimized complementary learning systems (Go-CLS) predicts that unpredictable memory components do not consolidate. A signal-to-noise ratio controls the extent of memory transfer

$$\lambda = \frac{\langle (\bar{W} \cdot X)^2 \rangle}{\langle (\epsilon)^2 \rangle}$$

λ : degree of predictability



Hippocampus is required to recall fully detailed episodic memories containing unpredictable elements.

Neocortex integrates memories to extract the predictable semantic content.